Prenatal Iodine, Maternal T4, and Neurodevelopment

Severe early iodine deficiency can disrupt neurodevelopment via maternal T4; late correction does not rewind early corticogenesis. This is not a brief for supplement maximalism in replete populations.

GRADE: HIGH-SEVERE; LOW-MILD-REPLETE | STATUS: DRAFT | DOMAIN: PREGNANCY | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Before mid-gestation the fetal brain depends on maternal T4. Severe iodine deficiency causes endemic cretinism (neurological injury that later iodine does not fully rewind). Milder early hypothyroxinemia is associated with altered cortical organization in experimental and some human data; that is not a license for maximal supplementation in iodine-replete settings.

EVIDENCE AND CAUSAL ASSESSMENT

Zimmermann et al. (2008) review iodine-deficiency disorders: preventable mental impairment at the severe end; salt iodization as a population measure. Morreale de Escobar et al. (2004) show maternal T4 reaches the fetal compartment early; locally generated T3 in cortex reaches adult values by mid-gestation; experimental early hypothyroxinemia disrupts neuronal migration.

EFFECT SIZE AND PERSISTENCE

Severe deficiency: large, lasting neurological injury. Mild deficiency: smaller and contested. Persistence of severe injury is structural; it is not “undone” by later correction of the same process.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (severe deficiency epidemiology and endocrine physiology, not an observational population phenotype association).

Population iodine status; over-translation of endemic data to replete settings; confounding of mild hypothyroxinemia with other pregnancy risks.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not justify supplement maximalism, or treating every learning difficulty as maternal iodine failure.

WHAT ACTION IS REASONABLY JUSTIFIED

Population iodization and preventing severe deficiency are justified. Extra iodine as cognitive enhancement in replete populations is not.

VERIFIED SOURCES & DOI IDENTIFIERS

- Zimmermann, Jooste & Pandav (2008), Lancet
- Morreale de Escobar, Obregón & Escobar del Rey (2004), Best Pract. Res. Clin. Endocrinol. Metab.